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VR Shopping, Fitting, and Recommendation Service Based on Interactive Interface

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- Online shopping malls that use **artificial intelligence technology** offer consumers quick and accurate access to suitable clothing through **personalized recommendation algorithms** and **virtual styling services**.
- VR shopping malls allow for the creation of **immersive store environments** without the use of physical resources, promoting environmental sustainability.
- Companies, in turn, utilize this technology to analyze consumer **purchasing and browsing patterns,** helping them improve their marketing strategies. Furthermore, consumer reviews and style sharing through social media platforms provide richer information about products and promote interaction among consumers, leading to increased purchase satisfaction and higher consumption
- **Speech recognition** is known as one of the most efficient means for facilitating communication between machines and humans. Both domestic and international retail companies are incorporating speech recognition technology into their businesses to offer consumers a more **convenient shopping experience**

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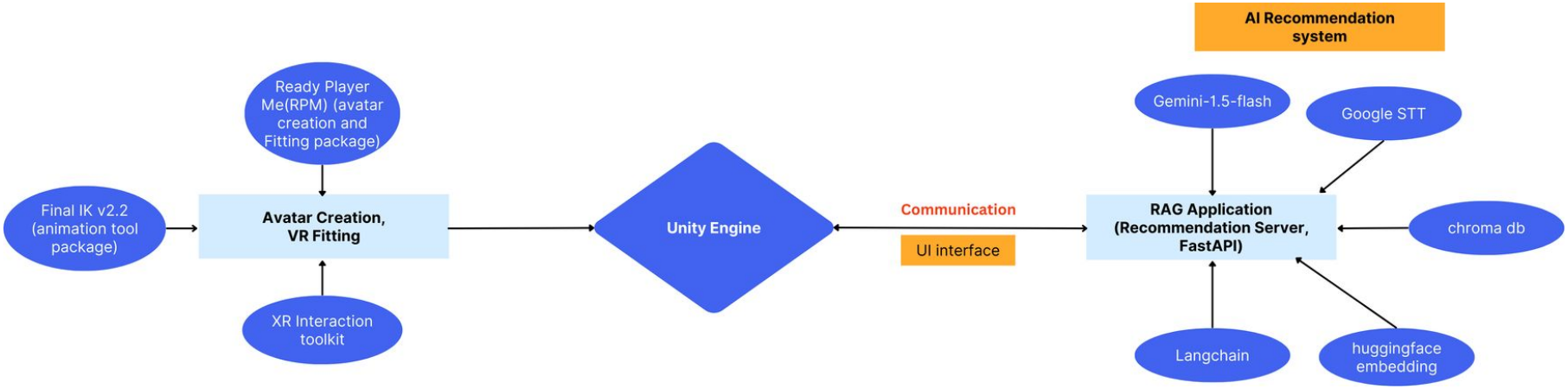
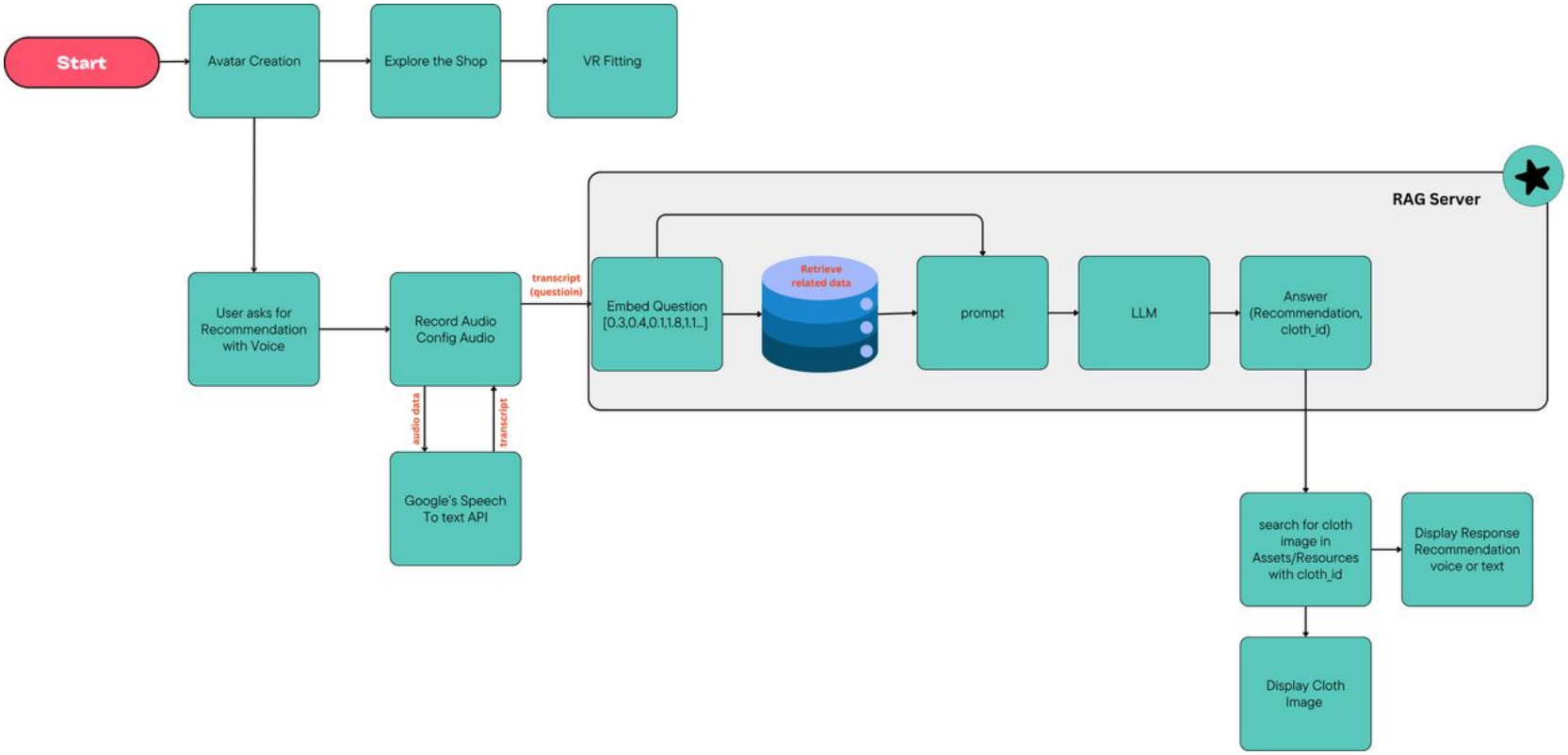
- The metaverse promotes user interaction, enables experience sharing, while **generative AI** has the advantage of providing **personalized services** tailored to user needs through **conversations**.
- **RAG (Retrieval Augmented Generation)** is a technique that expands the knowledge of **LLMs (Large Language Models)** with additional data, helping the model generate more accurate and specific answers.
- This paper proposes a conversational **interface-based VR shopping mall** where users can communicate with an **AI agent via voice** and receive clothing recommendations tailored to their situation, using the conversational **AI model Gemini (1.5-flash)**
- The proposed system not only includes the **AI-based clothing recommendation** function but also allows users to create their own avatars, shop freely, and try on clothes.

System Configuration



- Unity integrates tools such as Ready Player Me for avatar creation and Final IK for realistic animations, with the XR Interaction Toolkit providing smooth interactions.
- The AI recommendation system, powered by LangChain, Hugging Face embeddings, and ChromaDB, delivers personalized suggestions, supported by Google Speech-to-Text for voice interactions. This configuration combines VR and AI to enhance shopping in a seamless and engaging way.

System Process



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	Steps	Process
RAG Server	Step 1	RAG server starts embedding documents(shop info, Clothes details) (splitting and converting vector)
	Step 2	Store embedded docs in Vector Store
	Step 3	Response when called by unity server via API
Unity Server	Step 1	Unity server starts and configures all packages(RPM, Final IK, XR toolkit) and Google STT
	Step 2	User avatar creation and exploring VR shop and asking for **Recommendation** (API call to **RAG** server)
	Step 3	Display Response (voice or text)

Avatar Creation

Fitting

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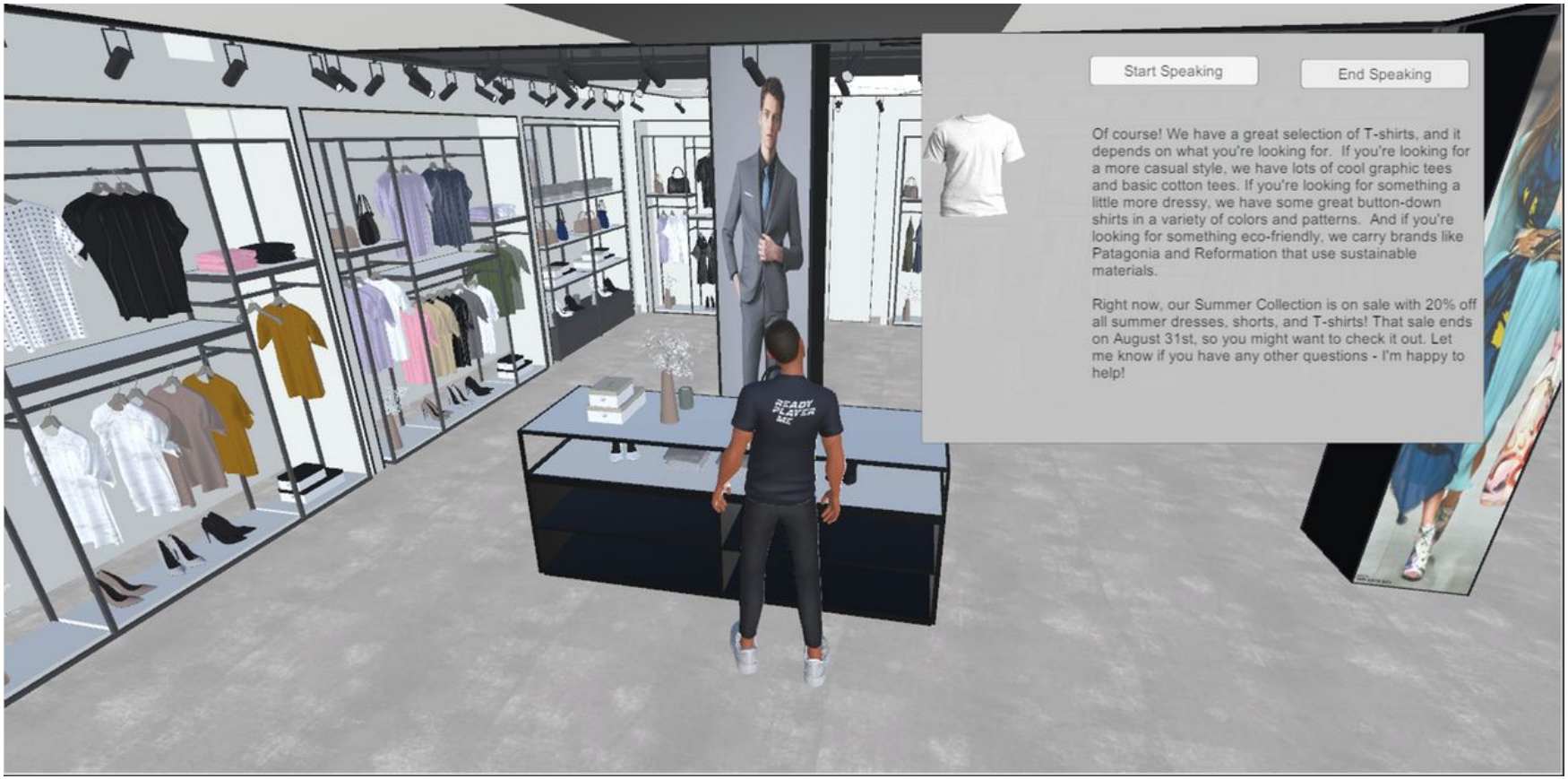
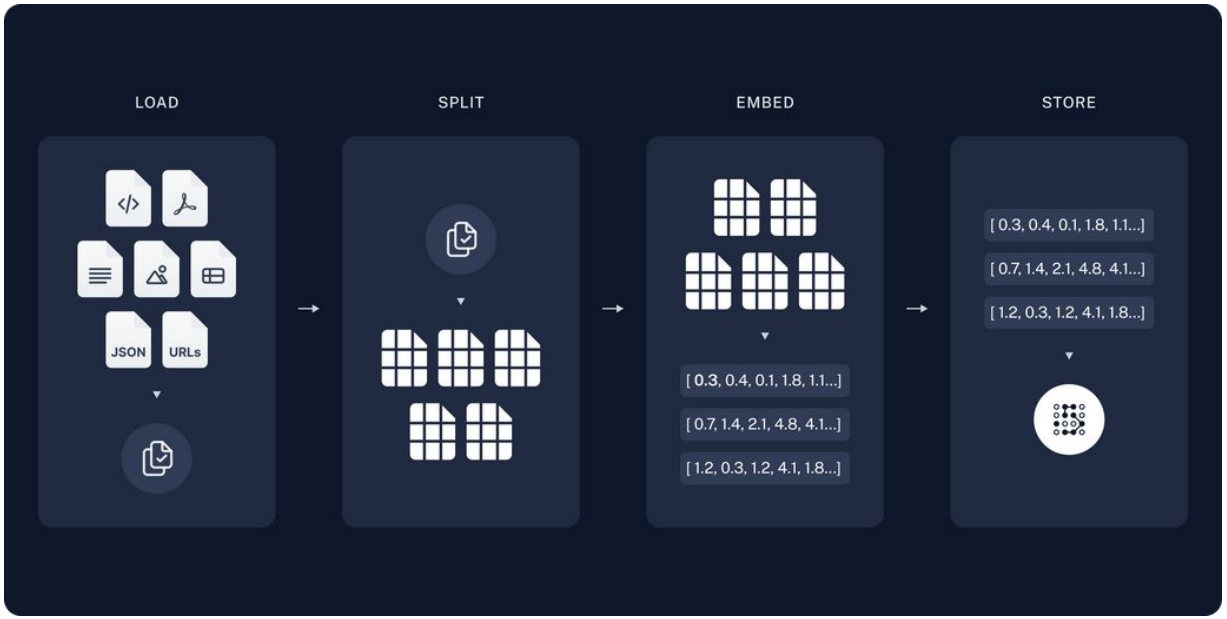


Recommendation

Embedding

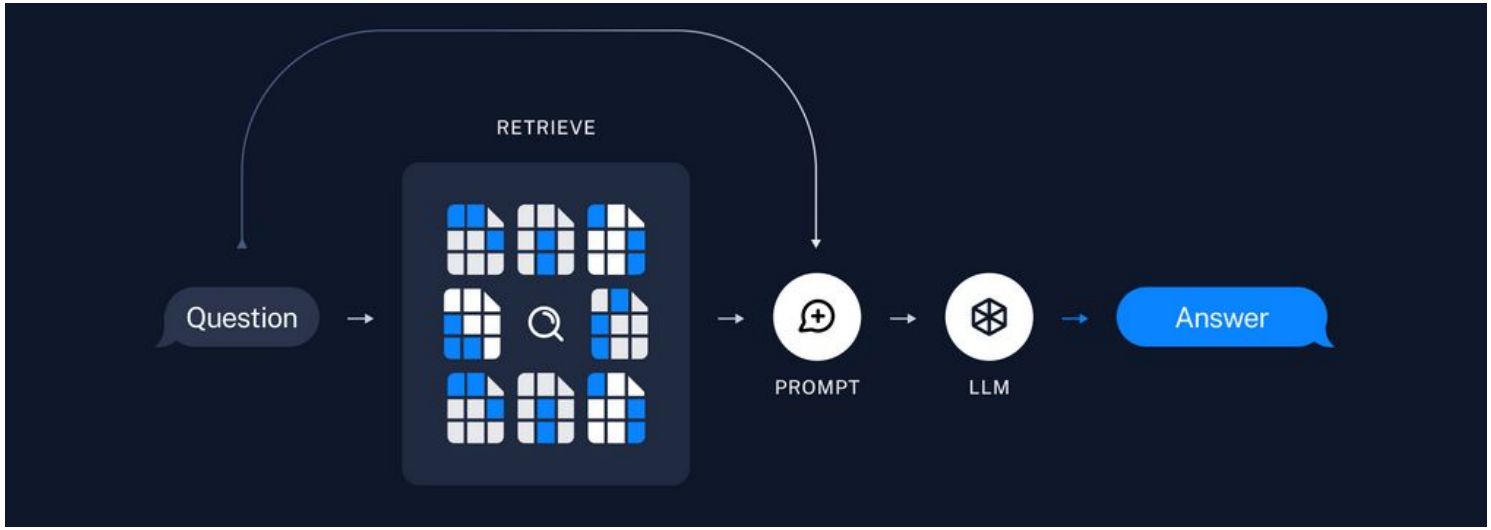


The Embedding begins with Loading documents such as text, PDFs, images, or JSON files.

The documents are divided into smaller chunks for efficient processing, then converted into semantic vector representations using an embedding model.

The embeddings generated are stored in a vector database optimized for similarity search, such as ChromaDB or Pinecone.

AI Recommendation System (RAG)



Retrieval Augmented Generation (RAG) system works to answer questions.

A user asks a question, and the system searches the most relevant pieces and used to create a detailed prompt, which is sent to a language model. The model then uses this information to generate a clear and accurate answer, which is delivered back to the user.

This process helps provide more reliable and context-aware answers by combining stored knowledge

- This study implemented a conversational interface-based virtual shopping mall by integrating the conversational AI model, Google Gemini, with a VR environment. The system's key features include automatic avatar creation and clothing fitting functionality using various libraries, as well as personalized clothing recommendation services based on the conversational AI model
- Through this research, it was demonstrated that the integration of metaverse platforms and generative AI provides a sense of familiarity and comfort for users, while also enhancing shopping satisfaction through personalized services.
- **In the future, we plan to create characters that closely resemble the actual customer's face and body size, and implement detailed size matching features.**
- A push notification feature will be developed to suggest clothing that matches items already stored in the user's virtual wardrobe when new products arrive, making it easier for users to make a purchase.

Thank You!